

University of Nottingham Malaysia

SCHOOL OF COMPUTER SCIENCE
A LEVEL 1 MODULE, SUMMER SEMESTER 2022-2023
RESIT

COMPUTER FUNDAMENTALS (COMP1027)

Time allowed: **ONE** Hour

Candidates may complete the front cover of their answer book and sign their desk card but must NOT write anything else until the start of the examination period is announced.

Answer ALL questions

Marks available for sections of questions are shown in brackets in the right-hand margin.

No calculators are permitted in this examination.

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn your examination paper over until instructed to do so.

ADDITIONAL MATERIAL:

None

INFORMATION FOR INVIGILATORS:

Please collect the exam paper at the end of the exam

1. Number Representation, Binary Arithmetic, Digital Logic & Boolean Algebra [30%]

- a. Given Table 1, calculate the 6 values labelled a-f. Show your calculation workings in detail and your representations for Binary, 1's Complement and 2's Complement must be in **8-bits width**. [12 marks]

Table 1.

Decimal	Binary	1's Complement	2's Complement
a	1111	b	c
f	e	0111 1010	d

- b. Given the following values of x, y, and z below (in Decimal):

- x = 110
- y = 89
- z = 126

Calculate values of x, y, and z in **8-bits width 2's Complement**. Show your working in detail. [6 marks]

- c. Using **8-bits TWO's complement**, calculate the following:

- i. $x - y$
- ii. $y - z$
- iii. $z - x$

Show your working in detail. Your answer should be in **8-bits width**. [6 marks]

- d. Given the **truth table** in Table 2 and Boolean Algebra in Figure 1:

Table 2.

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

Identity Name	AND Form	OR Form
Identity Law	$1x = x$	$0+x = x$
Null (or Dominance) Law	$0x = 0$	$1+x = 1$
Idempotent Law	$xx = x$	$x+x = x$
Inverse Law	$x\bar{x} = 0$	$x+\bar{x} = 1$
Commutative Law	$xy = yx$	$x+y = y+x$
Associative Law	$(xy)z = x(yz)$	$(x+y)+z = x+(y+z)$
Distributive Law	$x+yz = (x+y)(x+z)$	$x(y+z) = xy+xz$
Absorption Law	$x(x+y) = x$	$x+xy = x$
DeMorgan's Law	$\overline{(xy)} = \bar{x} + \bar{y}$	$\overline{(x+y)} = \bar{x}\bar{y}$
Double Complement Law	$\bar{\bar{x}} = x$	

Extracted from: L. Null, L. and J. Lohr, The essentials of computer organisation and architecture, Jones & Bartlett Publishers, 2003 (Fig. 1.3, P.26)

Figure 1.

P.T.O

- i. Write out the **simplified** expression. [4 marks]
- ii. Based on your answer in 1d(i), draw the Logic Circuit Diagram to implement the circuit (using the relevant combinations of ONLY basic gates). [2 mark]

2. Adders and ALU [40%]

- a. Given Table 3 below, fill in **binary values** for columns C_{out} and S , respectively. [16 marks]

Table 3.

Inputs			Outputs	
A	B	C_{IN}	C_{OUT}	S
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

- b. Based on your **completed** Table 3, illustrate the circuitry representation and structure (e.g., inputs and outputs) [7 marks]
- c. Based on your circuitry illustration in 2b, name the **type** of adder. [1 mark]
- d. Explain the **significant** difference between a *data selector* and *data distributor*. [3 marks]

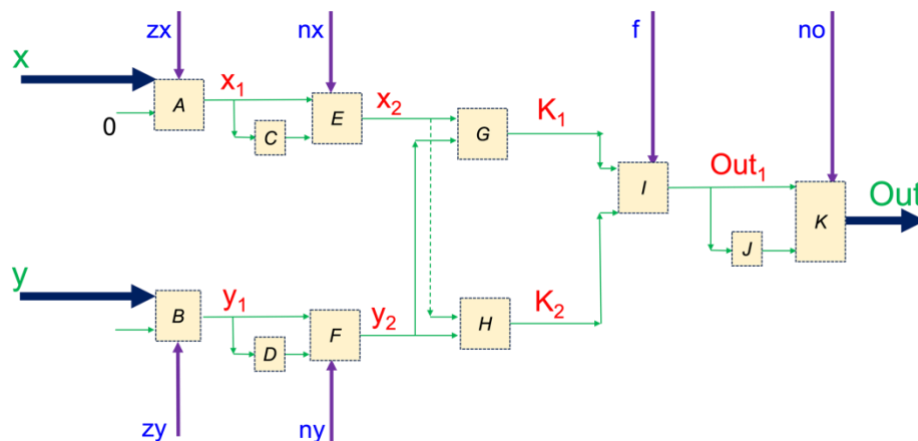


Figure 2

- e. Figure 2 illustrates an **incomplete** ALU. Please identify the missing components labelled A-K. [13 marks]

P.T.O

3. Machine Code & Networks [30%]

- a. A **Hack Computer** is a **16-bit** machine consisting of **five** fundamental elements. Name them and identify the respective components associated to each element.

[10 marks]

Table 4.

	Layer Name	Protocol Data Unit
7	-	
6	-	
5	-	
4	-	
3	-	
2	-	
1	-	

- b. Table 4 above shows the OSI Network stack, give the names for layers 1 till 7 and list each layer protocol data unit.

[14 marks]

- c. What is the **significant difference** between TCP/IP Model vs. OSI Model in terms of *layers*, *approach*, and *delivery*?

[6 marks]

END